A list of all protocols and their port numbers

Here is a list of common network protocols and their corresponding port numbers:

Well-Known Ports (0-1023)

1. **FTP (File Transfer Protocol)**
   - Port: 20 (Data Transfer), 21 (Control)
2. **SSH (Secure Shell)**
   - Port: 22
3. **Telnet**
   - Port: 23
4. **SMTP (Simple Mail Transfer Protocol)**
   - Port: 25
5. **DNS (Domain Name System)**
   - Port: 53
6. **HTTP (HyperText Transfer Protocol)**
   - Port: 80
7. **POP3 (Post Office Protocol version 3)**
   - Port: 110
8. **NNTP (Network News Transfer Protocol)**
   - Port: 119
9. **NTP (Network Time Protocol)**
   - Port: 123
10. **IMAP (Internet Message Access Protocol)**
    - Port: 143
11. **SNMP (Simple Network Management Protocol)**
    - Port: 161 (queries), 162 (traps)
12. **LDAP (Lightweight Directory Access Protocol)**
    - Port: 389
13. **HTTPS (HTTP Secure)**
    - Port: 443

14. **SMTPS (Simple Mail Transfer Protocol Secure)**

   - Port: 465

15. **FTPS (FTP Secure)**

   - Port: 989 (Data), 990 (Control)

Registered Ports (1024-49151)

1. **MySQL**

  - Port: 3306

2. **PostgreSQL**

  - Port: 5432

3. **RDP (Remote Desktop Protocol)**

  - Port: 3389

4. **Microsoft SQL Server**

  - Port: 1433

Dynamic and/or Private Ports (49152-65535)

These ports are used for ephemeral purposes and by various applications dynamically. There are no fixed protocols assigned to these ranges, as they are often allocated as needed by the OS or applications.

This is not an exhaustive list, but it covers many of the most commonly used protocols and ports.

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A list of all protocols and their port numbers in AWS

AWS services use a variety of protocols and port numbers. Here's a list of some commonly used AWS services and their associated protocols and port numbers:

Common AWS Service Ports

- **Amazon EC2**:

  - SSH: TCP 22

  - HTTP: TCP 80

  - HTTPS: TCP 443

  - RDP: TCP 3389
- **Amazon RDS**:
  - MySQL/MariaDB: TCP 3306
  - PostgreSQL: TCP 5432
  - Oracle: TCP 1521
  - SQL Server: TCP 1433
- **Amazon S3**:
  - HTTP: TCP 80
  - HTTPS: TCP 443
- **Amazon Redshift**:
  - TCP 5439
- **Amazon ElastiCache**:
  - Memcached: TCP 11211
  - Redis: TCP 6379
- **Amazon Elasticsearch Service**:
  - HTTP: TCP 80
  - HTTPS: TCP 443
- **Amazon MQ**messaging queue:
  - ActiveMQ: TCP 61616
- **Amazon WorkSpaces**:
  - PCoIP: TCP/UDP 4172
  - WorkSpaces client: TCP 443
- **AWS Lambda (for VPC traffic)**:
  - Varies depending on the service it interacts with, typically HTTP/HTTPS over TCP 80/443

AWS Management and API Access

- **AWS CLI/SDK**:
  - HTTPS: TCP 443
- **AWS Management Console**:
  - HTTPS: TCP 443

Other AWS Services

- **Amazon CloudFront**:
  - HTTP: TCP 80
  - HTTPS: TCP 443
- **AWS CodeCommit**:
  - SSH: TCP 22
  - HTTPS: TCP 443
- **AWS CodeDeploy**:
  - HTTPS: TCP 443
- **AWS CodePipeline**:
  - HTTPS: TCP 443
- **AWS CodeBuild**:
  - HTTPS: TCP 443
- **Amazon EMR**:
  - Varies depending on the application and Hadoop distribution used

This is not an exhaustive list, and AWS services may use additional ports and protocols based on specific configurations and use cases. For detailed information, it's best to refer to the official AWS documentation or the service-specific documentation.

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A list of all protocols and their port numbers in GCP

Here is a list of commonly used protocols and their port numbers in Google Cloud Platform (GCP):

Common Protocols and Ports

- **HTTP:** 80
- **HTTPS:** 443
- **SSH:** 22
- **RDP:** 3389
- **FTP:** 21 (Control), 20 (Data)

- **SMTP:** 25, 587, 465
- **IMAP:** 143, 993 (IMAP over SSL)
- **POP3:** 110, 995 (POP3 over SSL)
- **DNS:** 53 (TCP/UDP)
- **MySQL:** 3306
- **PostgreSQL:** 5432
- **MongoDB:** 27017
- **Redis:** 6379
- **Elasticsearch:** 9200 (HTTP), 9300 (Transport)
- **LDAP:** 389, 636 (LDAP over SSL)

Other Protocols and Ports

- **Telnet:** 23
- **TFTP:** 69 (UDP)
- **NTP:** 123 (UDP)
- **SNMP:** 161 (UDP)
- **NetBIOS:** 137-139
- **SMB:** 445
- **Syslog:** 514
- **Kerberos:** 88

Kubernetes Specific Ports

- **Kubernetes API Server:** 6443
- **kubelet:** 10250, 10255
- **kube-scheduler:** 10251
- **kube-controller-manager:** 10252
- **etcd:** 2379, 2380

This list covers the most commonly used protocols and ports, but there are many more depending on the specific services and applications you are using. You can refer to GCP's official documentation for more comprehensive and up-to-date information.

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A list of all protocols and their port numbers in Azure

Azure services use a variety of protocols and port numbers. Here's a list of some common Azure services and their associated ports:

1. **Azure Virtual Network**
   - **HTTP**: 80
   - **HTTPS**: 443
   - **DNS**: 53 (TCP/UDP)
   - **RDP**: 3389
   - **SSH**: 22

2. **Azure Storage**
   - **Blob Storage**: 80 (HTTP), 443 (HTTPS)
   - **File Storage**: 445 (SMB)
   - **Queue Storage**: 80 (HTTP), 443 (HTTPS)
   - **Table Storage**: 80 (HTTP), 443 (HTTPS)

3. **Azure SQL Database**
   - **SQL Database**: 1433

4. **Azure App Service**
   - **HTTP**: 80
   - **HTTPS**: 443

5. **Azure Kubernetes Service (AKS)**
   - **Kube API Server**: 443
   - **Kubelet**: 10250
   - **NodePort Services**: 30000-32767

6. **Azure Active Directory**

- **LDAP**: 389
- **LDAPS**: 636

7. **Azure Service Bus**
   - **AMQP**: 5671 (TLS), 5672 (non-TLS)
   - **HTTPS**: 443

8. **Azure IoT Hub**
   - **HTTPS**: 443
   - **MQTT**: 8883
   - **AMQP**: 5671

9. **Azure DevOps**
   - **HTTP**: 80
   - **HTTPS**: 443

10. **Azure Redis Cache**
    - **SSL**: 6380
    - **Non-SSL**: 6379

11. **Azure VPN Gateway**
    - **IKEv2 VPN**: 500 (IPSec), 4500 (IPSec NAT-T)
    - **OpenVPN**: 1194, 443

This is not an exhaustive list, but it covers many of the commonly used services in Azure. If you need specific port information for a service not listed here, please let me know!

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A list of all subnet protocols and their port numbers in AWS

In AWS, several subnet protocols are used to facilitate different types of network communications. Here is a list of some common protocols along with their associated port numbers:

1. **HTTP**: Port 80

2. **HTTPS**: Port 443

3. **SSH**: Port 22

4. **FTP**: Port 21

5. **FTPS**: Port 990

6. **SMTP**: Port 25

7. **IMAP**: Port 143

8. **IMAPS**: Port 993

9. **POP3**: Port 110

10. **POP3S**: Port 995

11. **DNS**: Port 53

12. **MySQL/Aurora**: Port 3306

13. **PostgreSQL**: Port 5432

14. **Microsoft SQL Server**: Port 1433

15. **Oracle Database**: Port 1521

16. **RDP (Remote Desktop Protocol)**: Port 3389

17. **LDAP**: Port 389

18. **LDAPS**: Port 636

19. **NFS**: Port 2049

20. **Elasticache (Redis)**: Port 6379

21. **Elasticache (Memcached)**: Port 11211

These ports and protocols are used in various AWS services and resources. When configuring security groups or network ACLs, it's important to specify the correct ports and protocols to ensure proper communication and security.

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A list of all subnet protocols and their port numbers in GCP

In Google Cloud Platform (GCP), similar to AWS, various protocols are used with specific port numbers. Here is a list of common protocols and their corresponding port numbers:

1. **HTTP**: Port 80

2. **HTTPS**: Port 443

3. **SSH**: Port 22

4. **FTP**: Port 21

5. **FTPS**: Port 990

6. **SMTP**: Port 25

7. **IMAP**: Port 143

8. **IMAPS**: Port 993

9. **POP3**: Port 110

10. **POP3S**: Port 995

11. **DNS**: Port 53

12. **MySQL**: Port 3306

13. **PostgreSQL**: Port 5432

14. **Microsoft SQL Server**: Port 1433

15. **Oracle Database**: Port 1521

16. **RDP (Remote Desktop Protocol)**: Port 3389

17. **LDAP**: Port 389

18. **LDAPS**: Port 636

19. **NFS**: Port 2049

20. **Redis**: Port 6379

21. **Memcached**: Port 11211

When configuring firewall rules in GCP, these ports and protocols are specified to allow or deny traffic to and from your resources. Make sure to configure them correctly to ensure proper communication and security.

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A list of all subnet protocols and their port numbers in Azure

In Microsoft Azure, similar to AWS and GCP, various protocols use specific port numbers. Here is a list of common protocols and their corresponding port numbers:

1. **HTTP**: Port 80

2. **HTTPS**: Port 443

3. **SSH**: Port 22

4. **FTP**: Port 21

5. **FTPS**: Port 990

6. **SMTP**: Port 25

7. **IMAP**: Port 143

8. **IMAPS**: Port 993

9. **POP3**: Port 110

10. **POP3S**: Port 995

11. **DNS**: Port 53

12. **MySQL**: Port 3306

13. **PostgreSQL**: Port 5432

14. **Microsoft SQL Server**: Port 1433

15. **Oracle Database**: Port 1521

16. **RDP (Remote Desktop Protocol)**: Port 3389

17. **LDAP**: Port 389

18. **LDAPS**: Port 636

19. **NFS**: Port 2049

20. **Redis**: Port 6379

21. **Memcached**: Port 11211

When configuring Network Security Groups (NSGs) or Azure Firewall rules in Azure, these ports and protocols are used to control traffic to and from your resources. Proper configuration is essential for ensuring security and enabling necessary communication.

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A list of all protocols and their port numbers in docker

In Docker, containers can use various network protocols and ports for communication. Here is a list of common network protocols and their typical port numbers used in Docker:

Commonly Used Protocols and Ports

1. **HTTP (HyperText Transfer Protocol)**
   - **Port:** 80 (default)
2. **HTTPS (HyperText Transfer Protocol Secure)**
   - **Port:** 443 (default)
3. **SSH (Secure Shell)**
   - **Port:** 22 (default)
4. **FTP (File Transfer Protocol)**
   - **Port:** 21 (default)
5. **SFTP (SSH File Transfer Protocol)**
   - **Port:** 22 (default, same as SSH)
6. **SMTP (Simple Mail Transfer Protocol)**
   - **Port:** 25 (default)
7. **IMAP (Internet Message Access Protocol)**
   - **Port:** 143 (default)
8. **IMAPS (IMAP over SSL)**
   - **Port:** 993 (default)
9. **POP3 (Post Office Protocol v3)**
   - **Port:** 110 (default)
10. **POP3S (POP3 over SSL)**
    - **Port:** 995 (default)
11. **DNS (Domain Name System)**
    - **Port:** 53 (default)
12. **MySQL**
    - **Port:** 3306 (default)
13. **PostgreSQL**
    - **Port:** 5432 (default)
14. **MongoDB**
    - **Port:** 27017 (default)
15. **Redis**
    - **Port:** 6379 (default)

16. **Elasticsearch**
   - **Port:** 9200 (default for HTTP), 9300 (default for TCP)
17. **RabbitMQ**
   - **Port:** 5672 (AMQP), 15672 (management interface)
18. **Kafka**
   - **Port:** 9092 (default)
19. **Nginx**
   - **Port:** 80 (default for HTTP), 443 (default for HTTPS)
20. **Tomcat**
   - **Port:** 8080 (default)
21. **Cassandra**
   - **Port:** 9042 (default)

Custom Ports

In Docker, you can map container ports to different host ports. For example, you can run a container with MySQL on port 3306 inside the container but map it to port 3307 on the host machine:

```bash
docker run -p 3307:3306 mysql
```

Protocol-Specific Ports

- **TCP (Transmission Control Protocol):** Used for most protocols listed above.
- **UDP (User Datagram Protocol):** Commonly used for DNS (port 53), DHCP, and some streaming services.

To list all open ports and their corresponding services in a Docker container, you can use the following command:

```bash
docker ps --format "table {{.ID}}\t{{.Names}}\t{{.Ports}}"
```

This command will display a table of running containers along with their port mappings.

--

A list of all protocols and their port numbers in Kubernetes

In Kubernetes, services are exposed using various protocols and port numbers. Here is a comprehensive list of commonly used protocols and their typical port numbers in Kubernetes:

Commonly Used Protocols and Ports

1. **HTTP (HyperText Transfer Protocol)**
   - **Port:** 80 (default)
2. **HTTPS (HyperText Transfer Protocol Secure)**
   - **Port:** 443 (default)
3. **SSH (Secure Shell)**
   - **Port:** 22 (default)
4. **FTP (File Transfer Protocol)**
   - **Port:** 21 (default)
5. **SFTP (SSH File Transfer Protocol)**
   - **Port:** 22 (default, same as SSH)
6. **SMTP (Simple Mail Transfer Protocol)**
   - **Port:** 25 (default)
7. **IMAP (Internet Message Access Protocol)**
   - **Port:** 143 (default)
8. **IMAPS (IMAP over SSL)**
   - **Port:** 993 (default)
9. **POP3 (Post Office Protocol v3)**
   - **Port:** 110 (default)
10. **POP3S (POP3 over SSL)**

- **Port:** 995 (default)

11. **DNS (Domain Name System)**
    - **Port:** 53 (default)
12. **MySQL**
    - **Port:** 3306 (default)
13. **PostgreSQL**
    - **Port:** 5432 (default)
14. **MongoDB**
    - **Port:** 27017 (default)
15. **Redis**
    - **Port:** 6379 (default)
16. **Elasticsearch**
    - **Port:** 9200 (default for HTTP), 9300 (default for TCP)
17. **RabbitMQ**
    - **Port:** 5672 (AMQP), 15672 (management interface)
18. **Kafka**
    - **Port:** 9092 (default)
19. **Nginx**
    - **Port:** 80 (default for HTTP), 443 (default for HTTPS)
20. **Tomcat**
    - **Port:** 8080 (default)
21. **Cassandra**
    - **Port:** 9042 (default)

Kubernetes-Specific Ports

1. **Kubernetes API Server**
   - **Port:** 6443 (default)
2. **Kubelet API**
   - **Port:** 10250 (default)

3. **Kube-scheduler**

   - **Port:** 10251 (default)

4. **Kube-controller-manager**

   - **Port:** 10252 (default)

5. **etcd**

   - **Port:** 2379 (default client port), 2380 (default peer port)

6. **CoreDNS**

   - **Port:** 53 (default for DNS), 9153 (metrics)

Protocol-Specific Ports

- **TCP (Transmission Control Protocol):** Used for most protocols listed above.

- **UDP (User Datagram Protocol):** Commonly used for DNS (port 53), DHCP, and some streaming services.

Exposing Services in Kubernetes

In Kubernetes, you can expose services using different types of service resources, such as `ClusterIP`, `NodePort`, and `LoadBalancer`. Here's an example of how to expose a service on a specific port:

```yaml
apiVersion: v1
kind: Service
metadata:
  name: my-service
spec:
  selector:
    app: MyApp
  ports:
    - protocol: TCP
```

```
    port: 80        # Port number the service is exposed on within the cluster
    targetPort: 8080 # Port number the application is listening on within the pod
  type: ClusterIP      # Type of service
```

Listing Open Ports in Kubernetes

To list all open ports and their corresponding services in a Kubernetes cluster, you can use the following command:

```bash
kubectl get services --all-namespaces -o wide
```

This command will display a table of all services along with their port mappings across all namespaces.